

Mahmoud Kassem

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SUMMARY

Computer Science and Applied Mathematics student at NYU Abu Dhabi graduating May 2026. Experienced in building scalable distributed systems and cloud-native applications with expertise in Java, Python, and microservices architecture. Strong foundation in data structures, algorithms, and system design with hands-on experience delivering production-ready applications. Proven ability to tackle complex technical challenges through innovative solutions and operational excellence.

EDUCATION

New York University (NYU) – Abu Dhabi

Sep. 2022 – May 2026

B.S. in Computer Science & Applied Mathematics

GPA: 3.89/4.0

Related Coursework: *Data Structures, Algorithms, Software Engineering, Database Systems* **Major GPA: 3.93/4.0**

Computer Networks, Operating Systems, Object-Oriented Design, Machine Learning

Study Away: *Paris, France (Spring 2024), Berlin, Germany (Summer 2024), New York, USA (Fall 2024)*

Graduation Date: May 2026 | Available: June 2026

TECHNICAL SKILLS

Languages: Java, Python, C++/C, JavaScript, TypeScript, SQL, Bash/Shell Scripting

Backend & Cloud: FastAPI, Node.js/Express, REST APIs, Microservices, Docker, Azure, CI/CD

Databases: MySQL, PostgreSQL, MongoDB, SQLite, NoSQL, SQLAlchemy ORM, Database Optimization

Systems & Tools: Linux/Unix, Git/GitHub, Distributed Systems, HPC Clusters, Performance Profiling

Engineering Practices: Data Structures & Algorithms, System Design, Automated Testing, Code Review, Monitoring

EXPERIENCE

Monta AI

Sep. 2025 – Present

AI Engineer Intern

San Francisco, CA (Remote)

- Engineering production RAG system using FastAPI on Azure to convert complex construction data (BIM models, IFC files) into queryable knowledge bases, reducing document lookup time by 70%.
- Developed parsing pipelines using LlamaIndex and Ollama to extract structured logic from technical documents, mitigating hallucination risks in high-stakes compliance contexts.
- Implemented comprehensive error handling and monitoring for distributed API endpoints, enabling real-time issue detection across production environments.

Software Analysis and Developer Support Lab – NYU

April 2025 – August 2025

Software Engineering Research Intern

Abu Dhabi, UAE

- Built automated pipeline using GPT-4 to systematically inject 6 types of semantic errors into code, generating novel dataset of 1,217 labeled samples for ML model evaluation.
- Developed UniXcoder-based classifier achieving F1-score of 0.64, outperforming linear baselines by 11.7% and establishing new benchmark for logic-aware code analysis.
- Optimized distributed processing across HPC cluster infrastructure, reducing pipeline execution time by 40% through parallelization and memory-efficient batch processing.

CAMeL Lab – NYU

March 2024 – May 2025

Software Developer & NLP Research Assistant

Abu Dhabi, UAE

- Achieved 95% synchronization accuracy across 77,429 tokens by engineering systematic mapping pipeline to align four disparate legacy datasets into unified, machine-readable corpus.
- Architected 68-function Python preprocessing library handling specialized character sets, contributed to open-source [CAMeL Tools](#) library (500+ global users).
- Designed automated test suites and conducted code reviews, maintaining high code quality standards across the development lifecycle.

- Mentored 50+ students over 300+ hours in data structures, algorithms, and OOP using Java, Python, and C++, bridging theoretical concepts with implementation.
- Provided technical guidance on debugging methodologies, complexity analysis, and software engineering best practices.

PROJECTS

notSoSimpleDB - Custom Database System | *Java*

[GitHub](#)

- Implemented core database internals from scratch: storage management, B-tree indexing, query optimization, transaction handling (ACID), and concurrency control with lock management.
- Applied advanced data structures for efficient query processing, achieving sub-linear lookup times through carefully designed index structures.

Navi - AI-Powered Computer Navigation Assistant | *Python, Gemini AI, 11Labs*

[GitHub](#)

- Built multimodal AI tool at HackHarvard 2025 integrating Gemini Vision for real-time screenshot analysis, ElevenLabs TTS, and Porcupine wake-word detection for hands-free operation.
- Engineered multi-turn conversational system providing step-by-step voice guidance, designed for elderly users navigating computer interfaces.

VoucherFinder - AI-Powered Discount Aggregator | *Python, Flask, React, TypeScript, SQLite*

[GitHub](#)

- Led team of 3 through complete SDLC, delivering 2K+ LOC production application with JWT authentication, rate limiting, and comprehensive error handling.
- Designed normalized database schema with optimized queries and indexing strategies, reducing average query latency by 60%.

RentIt - Community Rental Platform | *JavaScript, MongoDB, Node.js, Express, React*

[GitHub](#)

- Architected full-stack application as Product Owner, implementing secure authentication, transaction handling, and real-time WebSocket features with 90%+ test coverage.
- Designed NoSQL schema with complex aggregation pipelines, implementing proper indexing for high-performance data retrieval at scale.
- Integrated external APIs with retry mechanisms, circuit breakers, and graceful degradation to ensure fault tolerance.

Budgetly - Financial Management Platform | *Python, FastAPI, SQLAlchemy, SQLite*

[GitHub](#)

- Built RESTful backend with relational database management, input validation, and automated testing demonstrating SOLID principles.
- Implemented structured logging and health checks for production monitoring and troubleshooting.

PUBLICATIONS

- **Kassem, M.**, Ribeiro, F., & Nadi, S. “An Automated Methodology for Generating Labeled Datasets of Semantic Errors in Code.” *Accepted at the International Conference on Software Engineering (ICSE), Workshop on Large Language Models for Code (LLM4Code)*.
- Saeed, M., **Kassem, M.**, Amjed, M., Habash, N., & Bondok, R. “Qamar: The Quranic Morpho-syntactic Annotation Resource.” (In Preparation).

AWARDS & LEADERSHIP

Conference Grant (\$1,500) to present at ICSE 2026 (LLM4Code Workshop)

Dec. 2025

2nd place, NYUAD 13th International Hackathon for Social Good (Quantum AI for UN SDGs)

Apr. 2025

Advancement Opportunity Grant (AED 4,400) to participate in HackHarvard Hackathon

Oct. 2025